

## MOTOR SIZING AND CALCULATION

### Key Parameters

- Total Weight (W): 4.35 kg (approximately 42.67 N)
  - Number of Motors (n): 4
  - Motor Torque: 3 Nm each (Total = 12 Nm)
  - Motor Speed: 30 RPM
  - Wheel Radius (r): 0.035 m (3.5 cm)
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### Torque Requirements Calculation

#### A. Rolling Resistance (Tr)

This is the force required to keep the robot moving on a flat surface.

$$F_{\text{rolling}} = C_{rr} \times m \times g \\ = 0.02 \times 4.35 \times 9.81 \approx 0.85 \text{ N}$$

Torque due to rolling resistance:

$$T_r = F_{\text{rolling}} \times r \\ = 0.85 \times 0.035 = \mathbf{0.0297 \text{ Nm}}$$

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#### B. Acceleration Torque (Ta)

This is the torque required to accelerate the robot from rest to full speed in 0.5 seconds.

Top Velocity (v):

$$v = (2\pi \times \text{RPM} \times r) / 60 \\ = (2 \times \pi \times 30 \times 0.035) / 60 \approx 0.11 \text{ m/s}$$

Acceleration (a):

$$a = v / t = 0.11 / 0.5 = 0.22 \text{ m/s}^2$$

Force required for acceleration:

$$F_{\text{accel}} = m \times a \\ = 4.35 \times 0.22 = 0.957 \text{ N}$$

Torque due to acceleration:

$$T_a = F_{\text{accel}} \times r \\ = 0.957 \times 0.035 = \mathbf{0.0335 \text{ Nm}}$$

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### **Total Torque vs Available Torque**

Applying a safety factor of 2:

$$\begin{aligned} T_{\text{total per motor}} &= ((T_r + T_a)) / \text{Number of Motors} \\ &= ((0.0297 + 0.0335) ) / 4 \\ &= \mathbf{0.0158 \text{ Nm per moto}} \end{aligned}$$

**Total avl torque per motor=3Nm**

**Conclusion : Torque avl > Torque Required Hence**